

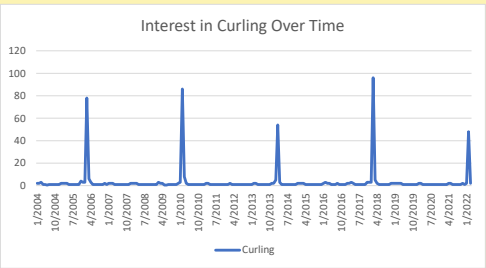
Project | Winter Sports Interest Analysis

- All of your answers should be typed or pasted into the grey boxes.
- Questions with grey answer boxes expect a specific answer. Once you input the correct answer, the question number box **will change from blue to green**.
- Questions with yellow answer boxes are free-response questions. There will be no change in the question number boxes (in blue), regardless of your answer.
- Hover your cursor over the blue hint box to get a helpful hint!

Part 1: Exploring Trends in Interest

1 Below, create a line chart that captures the trend in interest levels in "curling", over the time span recorded in the data.

Hint!



2 Summarize what the chart you created can tell us about search interest in Curling. Are there any repeating patterns? Where does this pattern come from?

The chart shows clear repeating spikes in search interest for Curling approximately every four years. These spikes occur during Winter Olympic years (2006, 2010, 2014, 2018), when Curling receives increased media coverage and global attention. Outside of Olympic years, search interest remains relatively low and stable.

3 Use ChatGPT to predict during which years or seasons search interest in Curling is likely to be highest.

Try this prompt: Based on historical search trends, during which years or seasons is interest in Curling likely to peak? What factors contribute to these spikes?

4 How does ChatGPT's answer align with your expectations and what you concluded from your visualization?

ChatGPT's response aligned exactly with my expectations and can conclude that people are most interested in curling when the Winter Olympics are ongoing.

5 Adjust your question to get more specific insights (e.g., narrow by year, country, or seasonality). What changed in the

When I narrowed the question to Olympic months (January–February) and Olympic years specifically, the response became more precise: it emphasized February of Olympic years as the clear peak window and gave stronger reasoning tied to Olympic scheduling and media coverage. The more specific prompt reduced vague seasonality talk and focused on the exact recurring spike periods shown in the chart.

6 How useful was ChatGPT in forming your hypothesis? Did you notice any biases or gaps?

ChatGPT was useful for turning the visual pattern into a clear hypothesis and for suggesting plausible drivers like media coverage and event timing. A gap is that it can sound confident without citing the dataset directly, and it may miss sport-specific real-world events unless prompted. It works best when paired with the chart evidence.

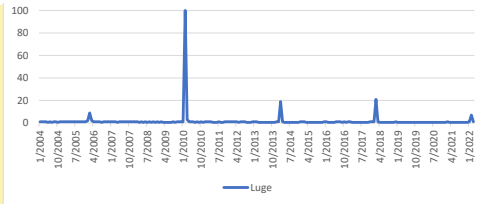
7 Below, create a line chart that captures the trend in interest levels in "luge", over the time span recorded in the data.

Hint!

Interest in Luge Over Time

120

Month	Curling	Luge	Speed Skating	Nordic combined	Biathlon	Alpine skiing	Bobsleigh	Hockey	Snowboarding
1/2004	2	1	2	0.5	1	3	1	23	34
2/2004	2	1	2	0.5	1	3	1	24	26
3/2004	3	1	1	0	1	1	1	23	15
4/2004	1	1	1	0.5	0.5	0.5	0.5	16	8
5/2004	1	0.5	1	0	1	0.5	1	15	6
6/2004	0.5	1	1	0	0.5	0.5	0.5	12	5
7/2004	1	0.5	1	0	1	0.5	1	11	6
8/2004	1	1	1	0	1	1	1	14	8
9/2004	1	1	1	0	0.5	1	1	18	11
10/2004	1	0.5	1	0.5	0.5	1	1	17	18
11/2004	1	1	1	0.5	0.5	1	1	20	25
12/2004	1	1	1	0	0.5	3	1	20	30
1/2005	2	1	1	0	1	3	1	21	29
2/2005	2	1	1	0.5	0.5	2	1	22	21
3/2005	2	1	1	0	0.5	2	1	21	14
4/2005	2	1	1	0	0.5	1	1	12	7
5/2005	1	1	1	0.5	0.5	1	1	11	6
6/2005	1	1	1	0.5	0.5	0.5	0.5	9	4
7/2005	1	1	1	0	0.5	0.5	1	9	6
8/2005	1	1	1	0	0.5	1	1	12	8
9/2005	1	1	1	0	0.5	1	0.5	15	12
10/2005	1	1	1	0.5	0.5	1	1	18	17
11/2005	4	1	1	0.5	0.5	2	1	23	25
12/2005	3	1	2	0	0.5	2	1	23	39
1/2006	3	2	3	0.5	1	2	2	23	30
2/2006	78	9	16	1	4	6	12	26	45
3/2006	6	2	3	0.5	1	2	2	23	20
4/2006	3	1	1	0.5	0.5	1	1	14	7
5/2006	1	1	2	0	0.5	0.5	0.5	13	6
6/2006	1	1	1	0	0.5	0.5	1	10	4
7/2006	1	1	1	0.5	0.5	0.5	0.5	9	5
8/2006	1	0.5	1	0	0.5	0.5	0.5	11	8
9/2006	1	1	1	0	0.5	1	1	15	11
10/2006	1	1	1	0	0.5	1	0.5	19	16
11/2006	2	1	1	0	0.5	1	1	21	22
12/2006	1	1	1	0.5	0.5	2	1	24	29
1/2007	2	1	1	0.5	0.5	2	1	22	23
2/2007	2	1	1	0.5	0.5	2	1	24	21
3/2007	2	1	1	0	0.5	1	1	22	12
4/2007	1	0.5	1	0	0.5	0.5	1	15	6
5/2007	1	1	1	0	0.5	0.5	0.5	12	5
6/2007	1	1	1	0	0.5	0.5	0.5	10	4
7/2007	1	1	1	0.5	0.5	0.5	0.5	9	5
8/2007	1	1	1	0.5	0.5	0.5	0.5	11	7
9/2007	1	1	1	0	0.5	1	0.5	14	10
10/2007	1	1	1	0	0.5	1	0.5	18	14
11/2007	1	1	1	0.5	0.5	1	1	21	19
12/2007	2	1	1	0.5	0.5	2	1	22	28
1/2008	2	1	1	0	0.5	2	1	22	23
2/2008	2	1	1	0.5	0.5	2	1	23	18
3/2008	2	0.5	1	0.5	0.5	1	1	21	11
4/2008	1	1	1	0	0.5	0.5	0.5	15	5
5/2008	1	0.5	1	0.5	0.5	0.5	0.5	12	4
6/2008	1	1	1	0	0.5	0.5	0.5	10	4
7/2008	1	0.5	1	0	0.5	0.5	0.5	9	4
8/2008	1	1	1	0.5	0.5	0.5	1	11	6
9/2008	1	0.5	1	0.5	0.5	0.5	0.5	14	9
10/2008	1	1	1	0.5	0.5	1	0.5	17	13
11/2008	1	0.5	1	0.5	0.5	1	0.5	20	20
12/2008	1	1	1	0.5	0.5	1	0.5	19	28
1/2009	1	0.5	1	0.5	0.5	2	1	20	23
2/2009	3	1	1	0.5	0.5	1	1	20	18
3/2009	2	0.5	1	0.5	0.5	1	0.5	20	11



8

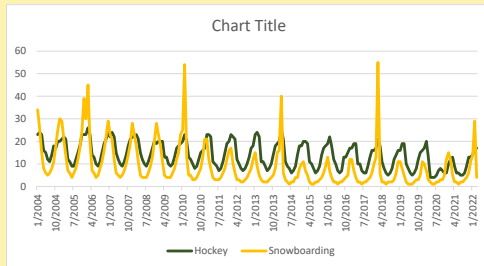
Compare this chart to the previous one. How does the overall search interest in Luge compare to Curling? Are there any notable spikes in interest? If so, what might explain these spikes? Use ChatGPT or Google to research significant events that could have influenced the trends you see.

Try this prompt: What major events could explain spikes in search interest for Luge in MM/YYYY?

Luge has a much lower baseline interest than Curling most of the time, but it had a major spike during the Winter Olympics February 2010 due to unfortunate death of Nodar Kumaritashvili. I used ChatGPT to gain the information on Mr. Kumaritashvili's passing.

9

Below, create a line chart that captures the trend in interest levels in "hockey" and "snowboarding", over the time span recorded in the data.



10

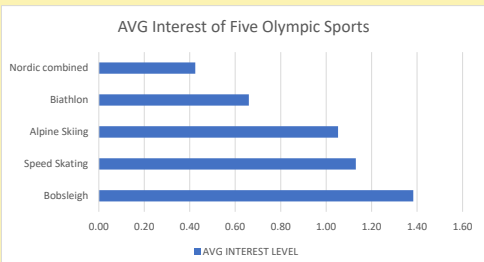
Describe the general trend in search interest for Hockey and Snowboarding based on the chart. How do they differ from the previous charts, and how are they similar?

Hockey stays relatively high across the entire timeline, with moderate bumps during winter and noticeable peaks around Olympic months, suggesting steady year-round interest. Snowboarding is also consistently popular but shows stronger seasonality, with higher interest in winter months and Olympic spikes. Compared to Curling and Luge, both Hockey and Snowboarding have much higher baseline interest rather than only spiking during the Olympics.

Try this prompt: How much does interest in "Ice Hockey" overlap with searches for "Hockey" during the Winter Olympics? Would relying on "Hockey" alone give an accurate picture of Olympic interest, or could it misrepresent trends?

11

Use the table in the **Explore the Five Sports** worksheet to create a bar chart that captures the average interest in the remaining sports that have not yet been plotted: Speed skating, Nordic combined, Biathlon, Alpine skiing, and Bobsleigh.



4/2009	2	0.5	1	0.5	0.5	0.5	0.5	13	5
5/2009	0.5	0.5	1	0.5	0.5	0.5	1	13	4
6/2009	0.5	0.5	1	0.5	0.5	0.5	1	10	4
7/2009	1	0.5	0.5	0.5	0.5	0.5	0.5	9	4
8/2009	1	1	1	0	0.5	0.5	0.5	10	6
9/2009	1	0.5	1	0	0.5	0.5	0.5	13	8
10/2009	1	1	1	0.5	0.5	1	1	17	12
11/2009	1	1	1	0.5	0.5	1	1	18	15
12/2009	2	1	2	0.5	1	2	1	19	23
1/2010	3	1	2	0.5	1	2	3	20	25
2/2010	86	100	19	3	6	10	19	23	54
3/2010	8	3	3	0.5	1	1	3	20	13
4/2010	2	1	1	0.5	0.5	0.5	1	13	5
5/2010	1	1	1	0.5	0.5	0.5	1	12	5
6/2010	1	1	0.5	0.5	0.5	0.5	0.5	10	3
7/2010	1	0.5	1	0.5	0.5	0.5	0.5	8	3
8/2010	1	1	0.5	0.5	0.5	0.5	0.5	9	4
9/2010	1	0.5	1	0.5	0.5	0.5	0.5	12	6
10/2010	1	1	1	0	0.5	1	1	15	8
11/2010	1	0.5	1	0.5	0.5	1	1	16	13
12/2010	1	1	1	0.5	1	1	1	17	21
1/2011	1	1	1	0.5	0.5	1	1	23	21
2/2011	2	1	1	0.5	1	1	1	23	12
3/2011	2	1	1	0.5	0.5	1	1	22	8
4/2011	1	0.5	0.5	0.5	0.5	0.5	1	11	4
5/2011	1	0.5	0.5	0	0.5	0.5	0.5	10	3
6/2011	1	0.5	0.5	0	0.5	0.5	1	9	3
7/2011	1	1	0.5	0.5	0.5	0.5	0.5	7	3
8/2011	1	0.5	0.5	0	0.5	0.5	1	8	4
9/2011	1	0.5	0.5	0.5	0.5	0.5	1	11	6
10/2011	1	1	0.5	0.5	0.5	0.5	1	15	8
11/2011	1	1	1	0	0.5	1	1	19	12
12/2011	1	1	1	0.5	0.5	1	1	20	16
1/2012	1	1	1	0.5	0.5	1	1	23	17
2/2012	2	1	1	0.5	0.5	1	1	22	10
3/2012	1	1	1	0.5	0.5	1	1	21	6
4/2012	1	1	0.5	0.5	0.5	0.5	1	12	3
5/2012	1	0.5	0.5	0.5	0.5	0.5	1	10	3
6/2012	1	1	0.5	0.5	0.5	0.5	1	8	2
7/2012	1	1	1	0.5	0.5	0.5	1	7	2
8/2012	1	1	1	0.5	0.5	0.5	1	8	3
9/2012	1	0.5	0.5	0	0.5	0.5	1	10	4
10/2012	1	0.5	0.5	0.5	0.5	0.5	1	13	6
11/2012	1	0.5	0.5	0.5	0.5	1	1	17	8
12/2012	1	1	1	0.5	0.5	1	1	18	12
1/2013	1	1	1	0.5	0.5	1	1	23	15
2/2013	2	1	1	0.5	0.5	1	1	24	9
3/2013	2	0.5	1	0.5	0.5	1	1	22	5
4/2013	1	0.5	0.5	0.5	0.5	0.5	1	11	3
5/2013	1	0.5	0.5	0.5	0.5	0.5	1	11	2
6/2013	1	0.5	0.5	0.5	0.5	0.5	1	9	2
7/2013	1	0.5	0.5	0.5	0.5	0.5	1	7	2
8/2013	1	0.5	0.5	0.5	0.5	0.5	1	8	3
9/2013	1	0.5	0.5	0.5	0.5	0.5	0.5	10	4
10/2013	1	0.5	0.5	0.5	0.5	0.5	1	14	5
11/2013	2	0.5	1	0.5	0.5	1	1	18	8
12/2013	2	1	1	0.5	0.5	1	1	19	15
1/2014	5	1	2	0.5	1	1	4	20	16
2/2014	54	19	17	2	7	8	41	25	40
3/2014	3	1	1	0.5	1	1	1	21	6
4/2014	1	0.5	0.5	0.5	0.5	0.5	1	11	3
5/2014	1	0.5	0.5	0.5	0.5	0.5	1	9	2
6/2014	1	0.5	0.5	0.5	0.5	0.5	0.5	8	1
7/2014	1	0.5	0.5	0.5	0.5	0.5	0.5	6	2

12	Which of the five Olympic sports plotted has the highest average search interest?	Bobsleigh

LevelUp		
13	Create a horizontal stacked bar chart of the search interest levels of speed skating, alpine skiing, Nordic combined, biathlon, and bobsleigh during February of the Olympics years. You should have the y-axis labels be the sport and the color be each Olympics game. See the LevelUp 1 Data1 worksheet for the data.	
14	What Sport had the highest total search interest of the time span in the dataset?	
15	In the Five Sports worksheet, re-create the table of average search interest values of the listed sports.	

8/2014	1	0.5	0.5	0.5	0.5	0.5	0.5	7	2
9/2014	1	0.5	0.5	0.5	0.5	0.5	1	9	4
10/2014	1	0.5	0.5	0.5	0.5	0.5	1	14	4
11/2014	1	0.5	1	0.5	0.5	1	1	18	8
12/2014	2	1	0.5	0.5	0.5	1	1	18	10
1/2015	2	1	1	0.5	1	1	1	20	11
2/2015	2	1	0.5	0.5	0.5	1	1	20	7
3/2015	2	0.5	0.5	0.5	1	1	0.5	19	5
4/2015	1	0.5	0.5	0.5	0.5	0.5	0.5	11	2
5/2015	1	0.5	0.5	0	0.5	0.5	0.5	10	1
6/2015	1	0.5	0.5	0	0.5	0.5	0.5	8	1
7/2015	1	0.5	0.5	0.5	0.5	0.5	0.5	6	2
8/2015	1	0.5	0.5	0.5	0.5	0.5	0.5	7	2
9/2015	1	0.5	0.5	0	0.5	0.5	0.5	9	3
10/2015	1	0.5	0.5	0.5	0.5	0.5	1	14	4
11/2015	1	0.5	0.5	0.5	0.5	1	0.5	17	6
12/2015	1	1	0.5	0.5	0.5	1	1	18	9
1/2016	2	1	1	0.5	0.5	1	1	19	13
2/2016	3	0.5	0.5	0.5	0.5	1	1	22	7
3/2016	2	0.5	0.5	0.5	0.5	1	1	18	4
4/2016	2	0.5	0.5	0.5	0.5	0.5	0.5	12	2
5/2016	1	0.5	0.5	0.5	0.5	0.5	0.5	10	2
6/2016	1	1	0.5	0.5	0.5	0.5	0.5	7	1
7/2016	1	1	0.5	0.5	0.5	0.5	1	6	2
8/2016	2	1	0.5	0.5	0.5	0.5	1	7	2
9/2016	1	0.5	0.5	0.5	0.5	0.5	0.5	13	3
10/2016	1	1	0.5	0.5	0.5	0.5	0.5	13	4
11/2016	1	0.5	0.5	0.5	0.5	1	0.5	15	5
12/2016	1	1	0.5	0.5	0.5	1	1	17	12
1/2017	2	1	0.5	0.5	0.5	1	1	17	12
2/2017	2	0.5	0.5	0.5	0.5	1	1	19	7
3/2017	3	0.5	0.5	0.5	0.5	1	0.5	19	4
4/2017	2	0.5	0.5	0.5	0.5	0.5	1	10	2
5/2017	1	0.5	0.5	0.5	0.5	0.5	1	10	2
6/2017	1	0.5	0.5	0.5	0.5	0.5	1	7	1
7/2017	1	0.5	0.5	0.5	0.5	0.5	1	6	2
8/2017	1	0.5	0.5	0.5	0.5	0.5	1	6	2
9/2017	1	0.5	0.5	0.5	0.5	0.5	0.5	9	3
10/2017	1	0.5	0.5	0.5	0.5	0.5	1	13	4
11/2017	3	1	0.5	0.5	0.5	1	1	16	6
12/2017	3	1	1	0.5	0.5	1	1	16	10
1/2018	3	1	2	0.5	1	2	3	17	13
2/2018	96	21	16	1	7	17	30	21	55
3/2018	5	1	1	0.5	0.5	1	1	18	6
4/2018	2	0.5	0.5	0.5	0.5	0.5	0.5	11	2
5/2018	1	0.5	0.5	0.5	0.5	0.5	0.5	8	2
6/2018	1	0.5	0.5	0.5	0.5	0.5	1	6	1
7/2018	1	0.5	0.5	0.5	0.5	0.5	1	5	2
8/2018	1	0.5	0.5	0.5	0.5	0.5	0.5	6	2
9/2018	1	0.5	0.5	0	0.5	0.5	0.5	8	2
10/2018	1	0.5	0.5	0.5	0.5	0.5	0.5	12	3
11/2018	2	0.5	0.5	0.5	0.5	1	0.5	14	6
12/2018	2	1	0.5	0.5	0.5	1	1	16	11
1/2019	2	0.5	0.5	0.5	0.5	1	1	16	11
2/2019	2	0.5	0.5	0.5	0.5	1	1	19	8
3/2019	2	0.5	0.5	0.5	0.5	1	1	19	5
4/2019	2	0.5	0.5	0.5	0.5	0.5	0.5	10	2
5/2019	1	0.5	0.5	0	0.5	0.5	0.5	8	1
6/2019	1	0.5	0.5	0.5	0.5	0.5	0.5	6	1
7/2019	1	0.5	0.5	0.5	0.5	0.5	0.5	5	1
8/2019	1	0.5	0.5	0.5	0.5	0.5	0.5	6	2
9/2019	1	0.5	0.5	0.5	0.5	0.5	0.5	8	3
10/2019	1	0.5	0.5	0.5	0.5	0.5	0.5	12	3
11/2019	1	0.5	0.5	0.5	0.5	1	0.5	15	5
12/2019	1	0.5	0.5	0.5	0.5	1	1	16	11
1/2020	2	0.5	0.5	0.5	0.5	1	1	17	10

2/2020	2	0.5	0.5	0.5	0.5	1	1	20	7
3/2020	1	0.5	0.5	0.5	0.5	0.5	1	13	3
4/2020	1	0.5	0.5	0.5	0.5	0.5	1	4	2
5/2020	1	0.5	0.5	0.5	0.5	0.5	1	4	1
6/2020	1	0.5	0.5	0.5	0.5	0.5	0.5	4	1
7/2020	1	0.5	0.5	0.5	0.5	0.5	0.5	5	2
8/2020	1	0.5	0.5	0.5	0.5	0.5	0.5	7	2
9/2020	1	0.5	0.5	0.5	0.5	0.5	0.5	8	3
10/2020	1	0.5	0.5	0.5	0.5	0.5	0.5	7	3
11/2020	1	0.5	0.5	0.5	0.5	1	0.5	6	5
12/2020	1	0.5	0.5	0.5	0.5	1	1	6	12
1/2021	1	0.5	0.5	0.5	0.5	1	1	10	15
2/2021	1	1	0.5	0.5	0.5	1	1	12	12
3/2021	1	0.5	0.5	0.5	0.5	1	0.5	13	5
4/2021	2	0.5	0.5	0.5	0.5	0.5	0.5	9	2
5/2021	2	0.5	0.5	0	0.5	0.5	0.5	6	2
6/2021	1	0.5	0.5	0.5	0.5	0.5	0.5	6	1
7/2021	1	0.5	0.5	0.5	0.5	0.5	1	5	2
8/2021	1	0.5	0.5	0.5	0.5	0.5	1	5	2
9/2021	1	0.5	0.5	0.5	0.5	0.5	0.5	6	3
10/2021	1	0.5	0.5	0.5	0.5	0.5	0.5	9	4
11/2021	2	0.5	0.5	0.5	0.5	1	1	13	6
12/2021	1	0.5	1	0.5	0.5	1	1	13	11
1/2022	2	1	1	0.5	0.5	1	2	14	14
2/2022	48	7	6	1	4	7	15	17	29
3/2022	2	1	0.5	0.5	0.5	1	0.5	17	4